

DATABASE DESIGN

INVENTORY SYSTEM FOR SCHOOLS

PROJECT A ICT30017

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1. Logical Database Design

This document explains the logical structure of the Daymap Inventory Management System database. It will contextualise the ER design by describing the purpose of each entity and justifying the relationships. This document will also explain how the schema supports the user stories and functional requirements defined by the client.

User story A

As a staff member, I want to view a list of all inventory items so that can see what resources are available.

User story B

As a staff member, I want to learn who loans out what items so that reports can be created about activity and resource use.

1.1 Entity Descriptions

Items

Items represent a generic resource type within the inventory of the school. This would include items such as “laptop”, “First Aid Kit” etc. This entity stores a advanced level of information such as name, description, stock count, and status. This supports User Story A. As it enables staff to view all available items and their current state and availability. This satisfies requirements relating to item metadata, categorisation, tagging and stock visibility.

ItemInstances

Iteminstances represent an individual physical unit of an item in the system. In this, each instance will have unique attributes such as serial numbers or expiry dates. This entity supports the requirements for items (per-unit metadata). It enables accurate tracking of availability, condition, and lifecycle of each unit.

Categories

This entity focuses on user-defined groupings that allow items to be organised into meaningful collections, such as ‘Textbooks’, ‘Equipment’, or ‘Medical Supplies’. Categories supports requirements for User story A.

Tags

Tags are labels used to classify and filter items and categories. Tags support requirements for User Story B.

ItemTags

This entity is a junction table linking items and tags. This supports requirements for User story B.

CategoryTags

A junction table linking Categories and Tags. This supports User Story B

Transactions

The transaction table functions as an append-only audit log that records every change to stock levels. It ensures full traceability of inventory activity by capturing events such as loans, returns, purchases, adjustments, and restocks. By storing who performed each action and which item or instance was affected, it provides the accountability and historical accuracy required for compliance and reporting. It directly supports User story B.

1.2 Relationship Explanations

Items <-> ItemInstances (1:M)

Each item can have multiple physical units. This relationship supports per-unit tracking, expiry management, and accurate stock representation. It satisfies requirements for both User Stories.

Items <-> Categories (M:N)

Items can belong to multiple categories, and categories can contain multiple items. This is implemented using the ItemCategories junction table.

Item <-> Tags (M:N)

Items can be assigned multiple tags for flexible filtering that is implemented using ItemTags.

Categories <-> Tags (M:N)

Categories can also be tagged, enabling hierarchical filtering and improved organisation. This is implemented using CategoryTags.

Items → Transactions

Transactions may optionally reference a specific instance. This supports user story mapping via enabling per-unit audit trails.

1.3 User Story Mapping

User story A

As a staff member, I want to view a list of all inventory items so that can see what resources are available.

Supported by:

- Items (Core item data, Status, Stock Count)
- ItemInstances (Availability of individual units)
- ItemCategories & Categories (Grouping + filtering)
- ItemTags & Tags (Tag-based filtering)
- Transactions (Loan, return history, effects on availability)

User Story B

As a staff member, I want to learn who loans out what items so that reports can be created about activity and resource use.

Supported by:

- ItemInstances (Which unit was loaned)
- Items (What type of item was loaned)
- Transactions (Loan, return history, effects on availability)

1.4 Requirements Validation

Requirements	Supported by
1.1 Unique item identifiers	Item.id
1.2 Item metadata	Items, Names, Description, Status
1.3 Categorisation	Categories, ItemCategories
1.4 Tagging	Tags, ItemTags, CategoryTags
2.1 Generic vs specific tasks	Items + Iteminstances
2.2 Per-unit metadata	ItemInstances, SerialNumber, ExpiryDate, MetaData
3.1 User-defined categories	Categories
3.2 Items in multiple categories	ItemCategories
3.3 Tags assignable to items/categories	ItemTags, CategoryTags
4.1 Stock tracking	Items.StockCount
4.3 Audit logging	Transactions
4.4 Append-only transaction history	Transaction (design decision)
7.1 Expiry tracking	ItemInstance.ExpiryDate

The mapping demonstrates that the Sprint 1 schema fully satisfies the functional requirements defined by the client.